

# **BLY 111:General Biology I**

**Topic: Interrelationship of Organisms**

**By**

**Dr. Gbenga F. Akomolafe**

**Department of Plant Science and  
Biotechnology**

**Federal University of Lafia**

# Objectives :

- At the end of this class, each student is expected to:
- Define the following terms: Symbiosis, commensalism, mutualism, parasitism and predation.
- Differentiate between the terms stated above
- Give examples under each term

# Types of Biological associations

- Symbiosis/ mutualism,
- commensalism,
- parasitism and
- predation.
- Competition
- Saprophytism

# Symbiosis

- The simple meaning of symbiosis is living together
- This is a close relationship between two organisms of different species in which both partners benefit.
- There is a close and often permanent contact between the two species;
- They cannot survive under natural conditions without each other. In other words, there is an obligatory contact between them

# Examples of Symbiotic relationships

- **Lichen:** This is an association made up of two organisms namely a fungus and a green alga living in close association together. The alga provides food and are enmeshed within the fungus.
- The fungi which encloses and protects algae from physical damage and also provides some of the water/moisture. The fungus benefits by using part of the food manufactured by the green algae.



- Nitrogen fixing bacteria in the root nodules of leguminous plants: *Rhizobium leguminosarium* living in the root nodules of leguminous plants obtain nutrients from the cells nitrogen requirement of the host plant, grow and multiply or reproduce there. In return rhizobium fixes nitrogen directly into the plant from the atmosphere, thereby increasing the nitrogen requirements of leguminous plants.
- Protozoans in the intestine of termites: protozoans help the termite to digest the cellulose in the food while the protozoa are protected by the termites.
- Ant-fungal association: Tropical ants have fungal gardens in their nests. This association helps in a quick natural decomposition of leaves. There is a proteolytic enzyme in the feaces of ants which helps in the digestion of protein within the fungi. The fungus is devoid of this enzyme.

- Bacteria in the rumen of ruminants: Bacteria and other protozoa in the rumen of ruminant animals like cattle, sheep and goat help the ruminant to digest cellulose to sugars, synthesize amino acids and vitamins from other substances while the ruminant in turn provide food and shelter for the bacteria.
- Human and bacteria: Fermenting bacteria living in the female reproductive organ produces acid that maintains a pH between 4.0 and 4.5, thereby suppressing the growth of yeast and other potentially harmful microorganisms while human provide medium for them to thrive.

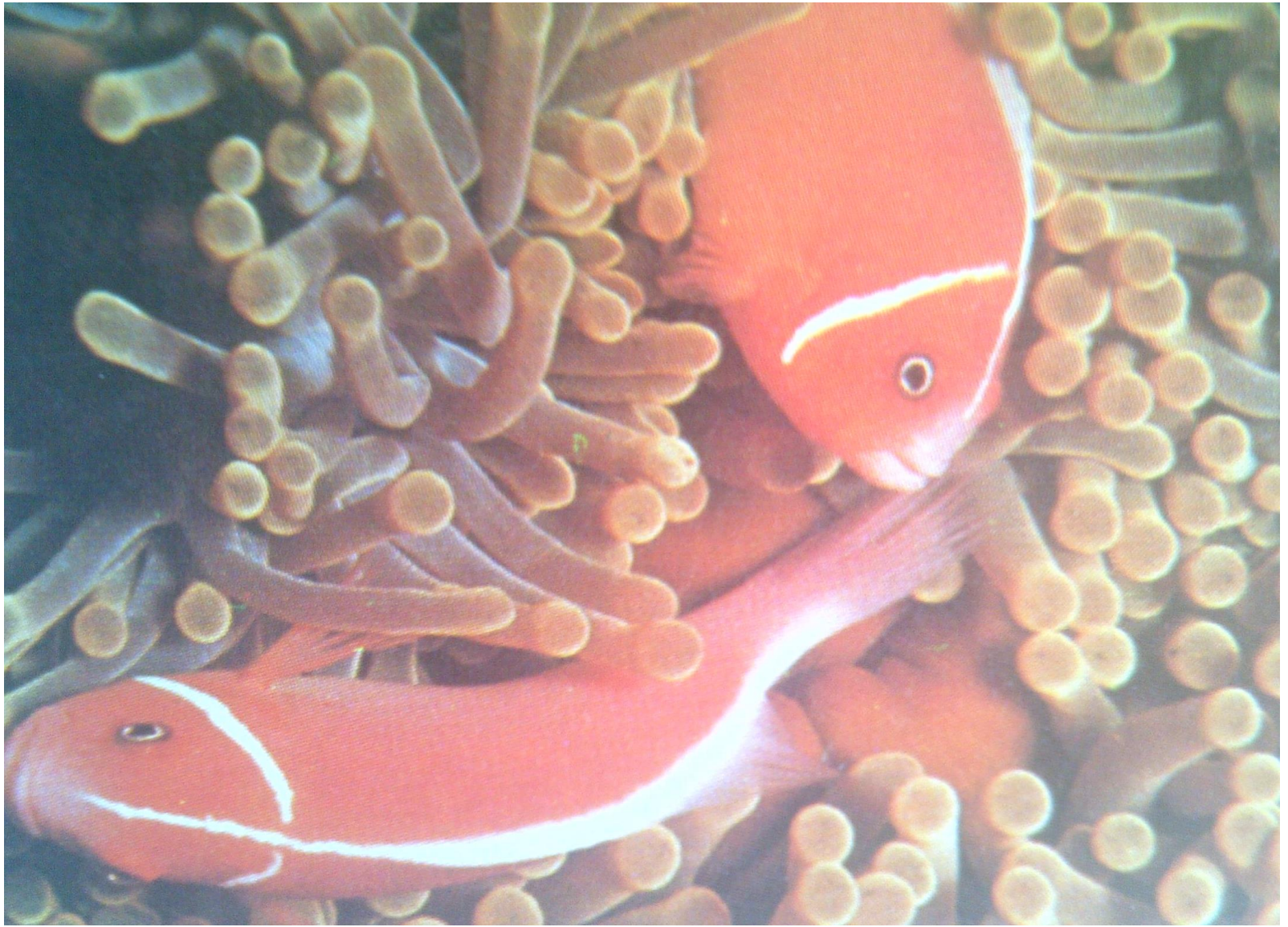


- Sea anemone and hermit crab: the sea anemone lives in the shell of the hermit crab, is carried around and also benefits from the bits of food that float away from the crab while feeding while the sea anemone does not offer anything to the crab.
- Cattle egrets and grazing cattle: the egrets concentrate their feeding around the grazing cattle while the grazing cattle raises grasshoppers feeding on grass.



- Clownfish and sea anemone: the clownfish lives close to a sea anemone and retreats among the anemone's tentacles whenever danger threatens. Within the anemone, the fish gains from it a covering of mucus that protects it from the anemone's stinging nematocysts. The fish derives protection from this relationship but the anemone also benefits because the clownfish attack other fish which come near including species which normally eat sea anemones





- The cleaner fish and customer: the cleaners gain a food source and the customers remain clean. The cleaner feeds on ectoparasites, bacteria and necrotic tissues from body surface of the customer.





- Flowers and insects: Insects obtain food from flowers in the form of pollen and nectars while in return, the insects bring about cross-pollination in the plant they visit, thus enabling plants to reproduce sexually







- Mycorrhiza: the nutrition of many higher vascular plants, e.g. Pinus species is directly tied up with the nutrition of higher fungi. The fungal mycelia get associated with the roots to form mycorrhiza. They may form a dense mantle either over the root surface, thus forming ectotrophic mycorrhiza or may inhibit the protoplast of roots to form endotrophic mycorrhiza. The hyphae obtain food and growth substances from the roots, while in turn provide water, nutrients and other compounds to the plant.



**Figure 13.11.** Mycorrhiza of pine (*Pinus sylvestris*). The swollen, much branched structure is the modified rootlet enveloped in a thick sheath of fungal tissue. (Courtesy

# Commensalism

- Commensalism is a symbiotic relationship that exists between two organisms living together in which one benefits from the association while the other neither benefits nor is harmed. Both partners are referred to as commensals. This is a positive-neutral relationship.

- Epiphytes: they grow on the body/branches of trees to be near light and their roots getting food from the soil in the crevices of the bark. They depend upon trees for support









- Remora Fish and Shark: The remora fish attaches itself to the body of the shark which carries it about. It feeds on the food particles left over by the shark, is protected and obtains shelter from the shark, while the shark is neither harmed nor benefits from the association.
- Man and intestinal bacteria: bacteria in the large intestine of man feed on digested food in there and receives food and protection from man whereas man neither gains nor suffers any disadvantage from the presence of the bacteria.

- Infectious disease and its carrier: the relationship between mosquito and the disease it carries. The mosquito is unaffected by the presence of the disease but it transfers the disease to the host in which the disease can reproduce or spread more easily to others.
- Bird nests in a tree: The bird is benefiting because the tree is giving the bird shelter and the tree is not getting anything in return.

# Parasitism

- This is a close association between two organisms in which one known as a parasite lives in or on and feeds at the expense of the other organism which is known as the host. The parasite benefits from the association while the host usually suffers or may die.



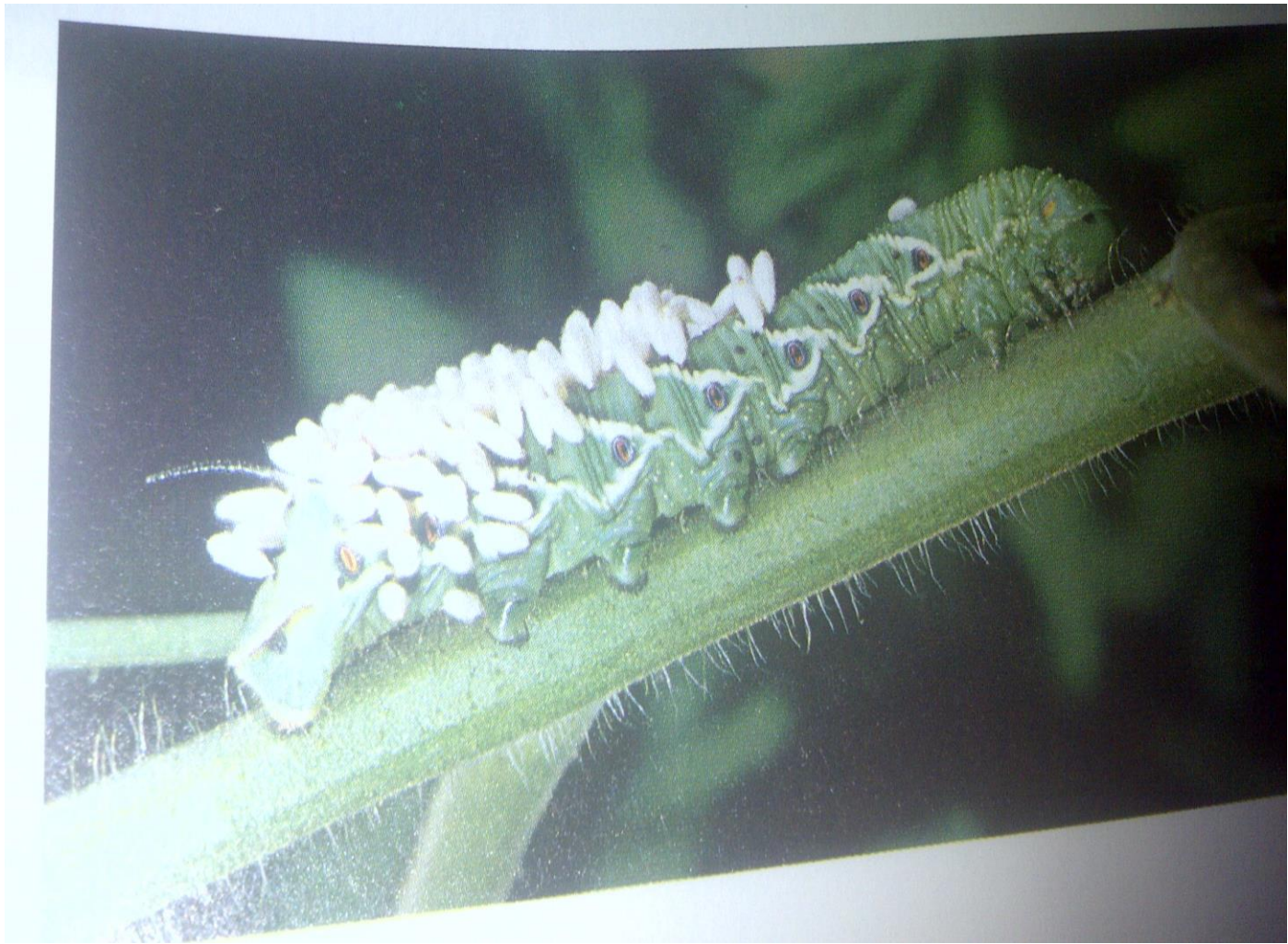
- Examples of parasites are viruses, bacteria that cause different diseases, certain protozoans that infect plants and animals, plant parasites such as mistletoe that draw nutrients from the branches of other plants.
- There are two types of parasites:
- Ectoparasites: These live on their hosts and do not enter the internal structure of the body system e.g. lice, ticks, fleas, mosquitoes, leeches, lamprey fish.
- Endoparasites: these are internal parasites that live inside the body of the host. E.g. malaria parasites and tapeworms.

- **Ectoparasites** examples:
- Man and mosquito: Mosquito feeds on man while transferring the disease called malaria.
- Lice that feeds on blood from human head
- The tsetse fly feeding on the blood of man and antelope. It is a vector of trypanosomes that causes African sickness

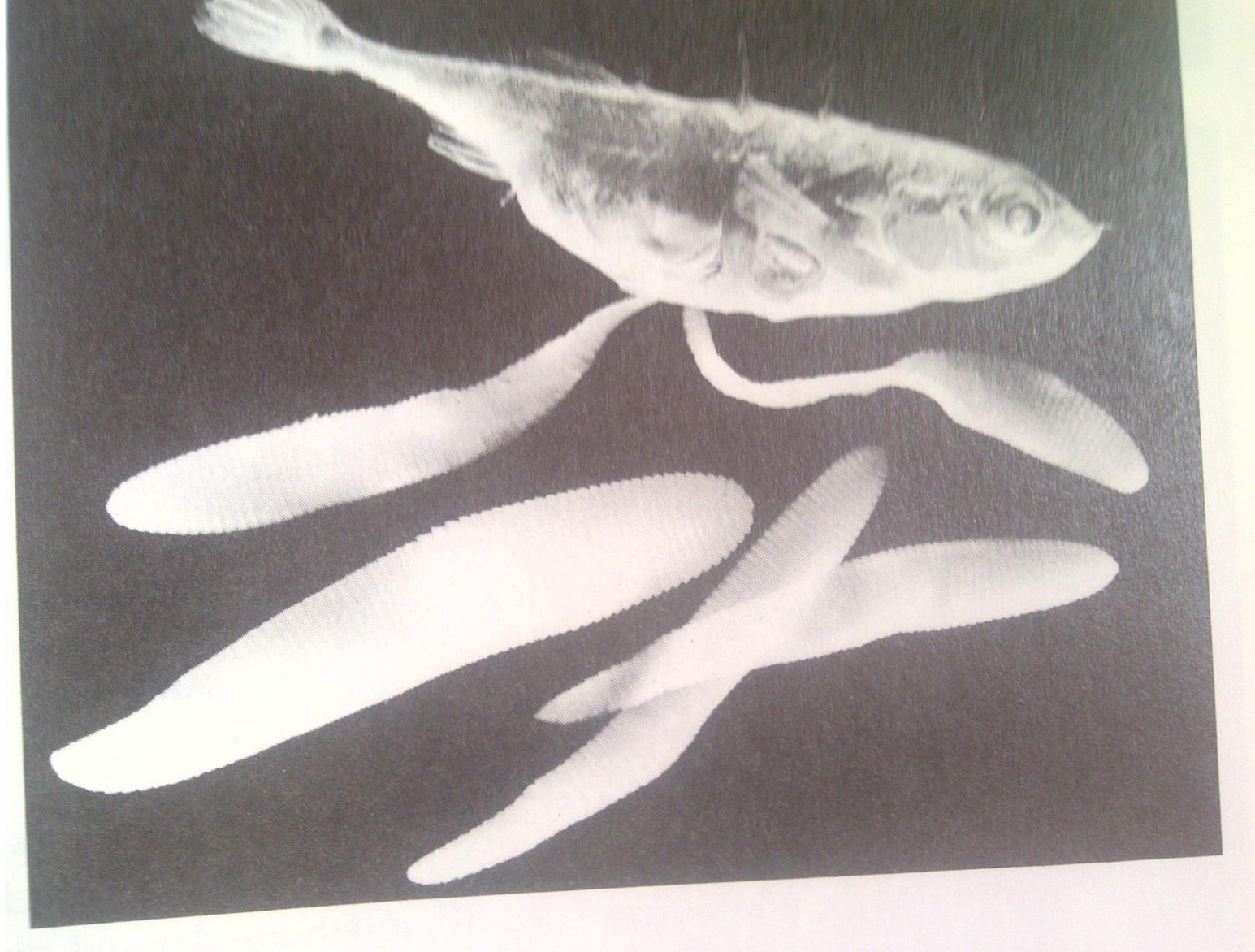


- Lamprey and fish: Certain species of lampres feed on the blood and body fluids of other fishes. When a suitable host is located, lamprey uses its disk-shaped mouth to attach to the host, scrapes a hole in the host and secretes a chemical that keeps the host blood from clotting. It then feeds on the blood and fluids that leak from the wound. The host may either recover or bleed to death from an infection.
- Eggs of wasp on tomato caterpillar
- Cestode parasites on fish





- Ticks and fleas that live off of a dog/cat/cattle/donkeys etc.
- Mistletoe and flowering plants: Mistletoe is a plant parasite that lives on other larger flowering plants. The host gives support and raises the mistletoe to a point where it can receive sunlight. The parasite also absorbs water and mineral salt from the host while the host suffers harm by losing to the parasite part of the water and mineral salts absorbed.







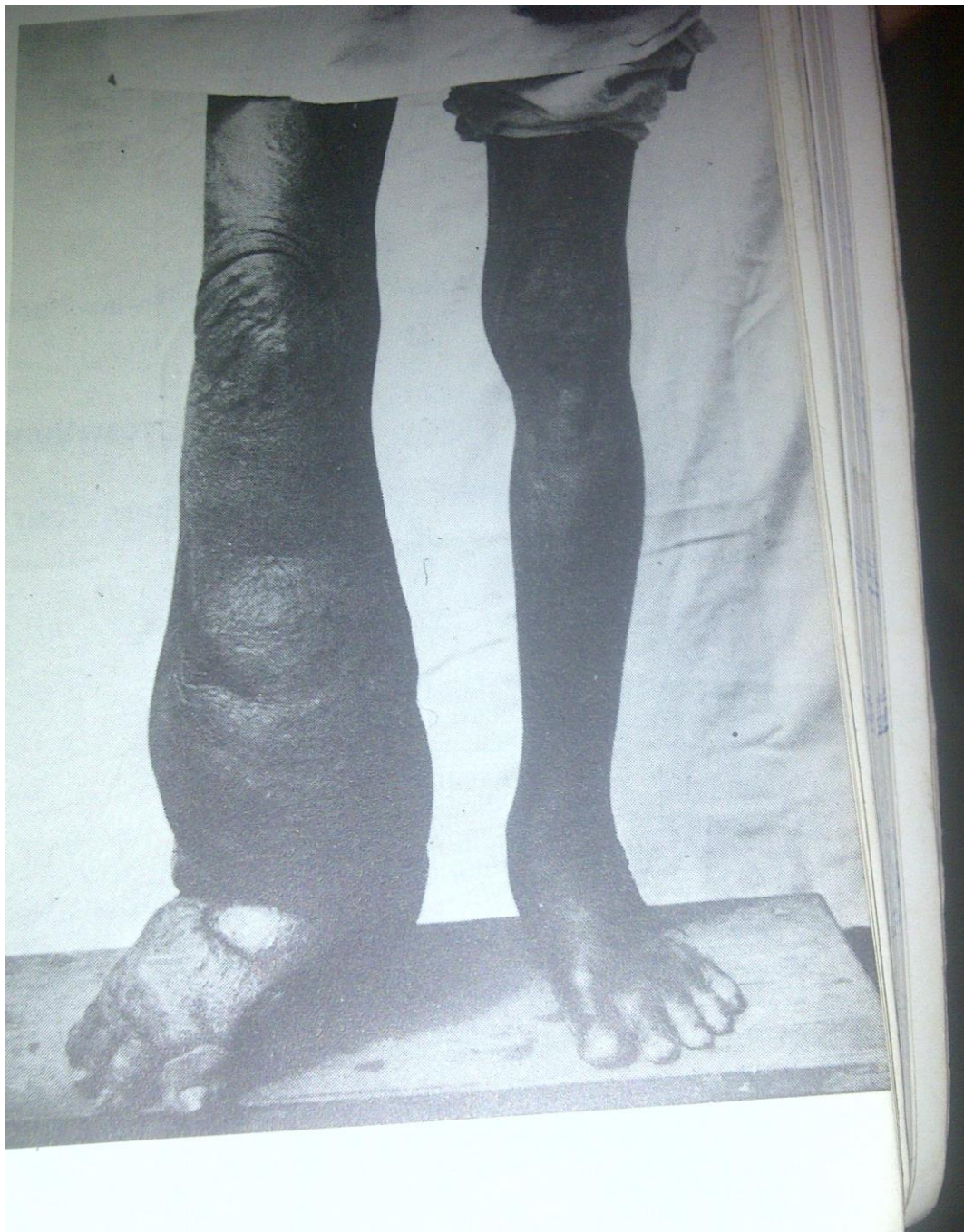


- Endoparasites:
- Man and the tapeworm: the tapeworm is a parasite that lives in the small intestine of man, obtains food, shelter and is protected from the digestive fluids in the intestine. The host, man suffers harm because he loses digested food to the tapeworm.
- *Ascaris* (nematode ) *Ascaris lumbricoides* is a large parasitic nematode that inhabits the human small intestine where it feeds on digested food.



- *Schistosoma* or the blood fluke: this inhabit and reproduce within the blood vessels of the intestine or urinary tract.
- Elephantitis caused by the filaria worm *Wuchereria bancrofti*, a parasitic nematode









- **Predation :**
- This is the type of relationship between two organisms in which the predator hunts and kills the other called the prey and directly feeds on it. The predator is usually larger in size and stronger than the prey. It benefits by deriving its food while it kills the prey. Predators are usually carnivores and an herbivore/plant relationship can also occur. Many predators have structures that enable them to find and capture prey such as claws, teeth, fangs, stingers and poisons

- Examples include:
- The hawks and chicks of domestic fowls: the predator here is the hawk that catches, kills and eats the prey (young chicks) of domestic fowls.
- The lion and the goat: the lion which is stronger and bigger than the goat kills and eats the prey (goat). The lion benefits while the goat is completely eliminated.
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- Rattle snakes and mammals: the rattle snake have an acute sense of smell with specialized heat sensitive-pits located below the nostril. This is used in locating and killing their prey such as birds and mammals by injecting them with toxin through their fangs.
- **Herbivore/Plant** relationship examples include:
  - Animals feeding on plants
  - Squirrel feeding on nuts



# Competition

- This involves the interaction among two organisms of the same or different species in which one outgrows the other and survives while the other neither grow nor survive.
- This is as a result of limited resources such as water, light, nutrients ,space, gases etc. during competition, one organism controls one or more of these resources which enables it to grow and survive while the other neither grow nor survive leading to elimination. This is a negative relationship because both organisms are harming each other.

- There are two types of competition:
- Intra-species competition: A competition between members of the same species
- Inter-species competition: a competition between members of different species
- Examples
- Flowering plants and grasses: flowering plants due to its size and numerous branches and leaves are capable of eliminating the grasses and depriving them of nutrients, space and light and this may eventually lead to their death.

- Domestic fowls and young chicks: domestic fowl deprive the young chicks of food which may eventually lead to the death of the young chicks.

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# Saprophytism

- Saprophytes draw their nourishment from dead plant and animal remains.
- They do not require light as a source of energy and hence can occur in deep shade or dark caves. They absorb nutrient from the decaying medium.
- They are free living forms and include bacteria and fungi.
- They are a vital link between the living and non-living components.





